

Problem-Sized Lean Worlds

From a Lean corpus to a short paper, open to outside contribution

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ABSTRACT

A correct Lean theorem can acquire a false meaning when its hypotheses disappear from a paper. Plectis records the passage from formal results to prose within a repository organised around individual problems. We follow a weighted-support criterion for reciprocal Mersenne series through its proof, its two registered Lean supports and its written explanation. The publication checks compare every registered support with the supplied evidence and detect changes to source bytes; the same records retain limitations and contributor credit through revision. A historical test rejected nine of ten false edits but accepted a false claim of completion. This case identifies the remaining task for authors and reviewers: deciding whether the recorded sources justify the words the reader sees.

1 Introduction

Consider a theorem asserting that $\sum_{a \in A} (2^a - 1)^{-1}$ is irrational under a summability condition on the set of exponents A (Section 3). Two faults in publishing this result require different checks. The paper might retain only one of its two registered Lean supports. Or it might retain both links but omit the summability hypothesis, thereby asserting a solution of the arbitrary-support problem, which remains open.

For each mathematical problem, Plectis keeps the question, prior sources, computations, proofs and unresolved steps together; we call this collection a problem world. It records which Lean results support a paper statement and which source passages explain it, so that a reviewer can make these comparisons at a specified revision. Figure 1 shows where these records enter: a mathematical claim is selected from the accumulated work, all its registered supports are compared with the supplied evidence, and an author turns it into a paper that readers may question or extend.

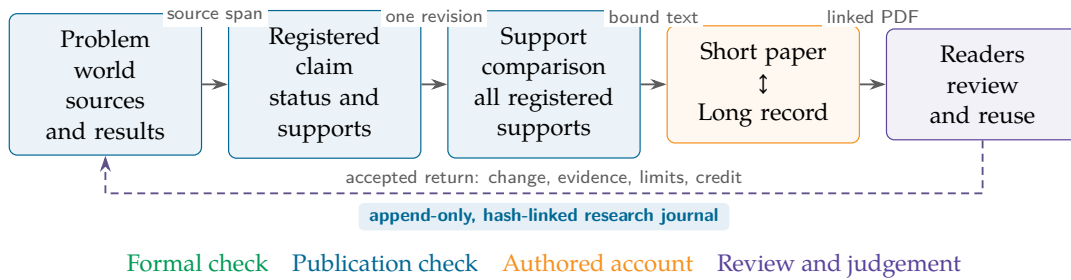


Figure 1: The life of one claim: arrows name the objects exchanged, and the support comparison checks the complete recorded set without executing a new formal proof; the short and long accounts remain linked.

* *Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Erdős Problems supplies the questions and their public discussion, while Lean and mathlib supply the formal language and library [1, 11, 76]. Formalisation blueprints link a written mathematical argument to named Lean declarations [35, 74]. Persistent problem records and open agent collaboration have antecedents in the Equational Theories Project and Prove2Me [75, 31]. The additional relation described here records a statement together with all its registered supports and the source passages used to explain it.

We first describe what a problem world lets a reader recover. The worked example then supplies the mathematical reason for its support hypothesis, before we follow the theorem into the publication checks, editorial revisions and contribution record. The historical false-edit test returns to the distinction that opened the paper.

2 A problem-sized Lean world

Within this collection, a reader may want a proof, a counterexample to a proposed extension, or the argument behind a failed approach. Exact experiments and corrected references remain beside the original question as contributions in their own right. The first task is to find the relevant work without compiling the whole repository. Four views answer different parts of that question (Figure 2).

A declaration is a named Lean definition, theorem or lemma. The declaration index locates its source; the dependency index records which earlier declarations the selected results use. A separate graph records authored interpretations of those declarations and their connections, while a claim registry records selected mathematical assertions, their status and the obligations still open.

ONE PROBLEM WORLD

Question and literature

Lean statements and proofs

Claims and open obligations

Computations and certificates

Failed approaches

Paper and supporting record

FOUR VIEWS, NO LEAN BUILD NEEDED

Declaration index: exhaustive source locations

Dependency index: exact edges for loaded roots

Semantic graph: 139,818 of 152,386 interpreted (91.8%)

Claim registry: 160 claims, eight statuses, 19 open obligations

68 factorial series 243 reciprocal tails 249 totient series 251 prime gaps

257 Mersenne subseries 269 running LCM 1041 lemniscates 1049 rational-base series

Corpus snapshot: 8 worlds, 1,903 Lean modules, 22 registered papers

Figure 2: A world contains the evidence and unresolved work for one question; the counts describe the whole recorded corpus, and exhaustive declaration locations do not imply exhaustive mathematical interpretation.

Starting from a recorded assertion, a reader can therefore locate its declarations, follow the recorded dependencies and request the specified lines of source needed for a particular question. The declaration index is exhaustive within its inventory, and the dependency edges are exact for

the selected starting declarations. The recorded interpretations are selective and largely contextual; they do not document an individual reading of every proof. A source link by itself records no successful execution. When a question requires a Lean check, the reader can request a focused build and inspect its recorded result.

3 A theorem and the reason for its hypothesis

For an infinite set A of positive integers and an integer $b \geq 2$, write

$$X_A(b) = \sum_{a \in A} \frac{1}{b^a - 1}.$$

The question is when a condition on the exponents forces this number, and the corresponding sum over every infinite subset of A , to be irrational.

The condition below rewards exponents divisible by large powers of a fixed collection of primes. For the prime 2 alone, the relevant part of $a = 2^k m$, with m odd, is 2^k . For a finite nonempty set P of primes, this becomes $h_P(a) = \prod_{p \in P} p^{v_p(a)}$.

The weighted-support theorem of the Problem 257 paper states that, if

$$W_{2,P}(A) := \sum_{a \in A} \frac{h_P(a)}{a(2^{h_P(a)} - 1)} < \infty, \quad (1)$$

then $X_B(b)$ is irrational for every integer $b \geq 2$ and every infinite $B \subseteq A$ [2, Theorem 1.2].

There is also a fixed-base version, with 2 replaced by b in the hypothesis. We explain the weighted condition and its proof first, then locate both registered Lean supports in Figure 3. Their recorded status does not report a new comparison of source revisions, Lean build or Comparator run. Comparator checks a selected Lean implementation against a separately stated challenge, as described in Section 4. This is a sufficient condition on the support; the question for an arbitrary infinite set of exponents in base two remains open.

Erdős had already stated that $\sum_{a \in A} 1/a < \infty$ suffices for this all-base irrationality conclusion [3, p. 222]. The source paper credits Duverney and Tachiya for an arithmetic-progression averaging argument of the kind used to select a small positive displacement [4, Section 2, (2.3)–(2.9)].

The distinction from the reciprocal-summability criterion is visible in the source paper's example

$$A_* = \{2^k m : k \geq 1, m \text{ odd}, 1 \leq m \leq 2^{2^k}\}, \quad P = \{2\}.$$

Within the layer $v_2(a) = k$, reciprocal summation discounts each odd m by 2^{-k} ; weighted summation discounts it by $(2^{2^k} - 1)^{-1}$.

To compare the layers, put $S_r = \sum_{1 \leq m \leq 2^r, m \text{ odd}} 1/m$. Grouping the odd integers into intervals $[2^j, 2^{j+1})$ gives $r/4 \leq S_r \leq r$ for integers $r \geq 2$. The reciprocal contribution of each disjoint layer is at least $2^{-k} S_{2^k} \geq 1/4$. Its weighted contribution satisfies

$$\frac{S_{2^k}}{2^{2^k} - 1} \leq \frac{2^k}{2^{2^k} - 1} \leq 2^{1-k},$$

so the layer contributions have a convergent geometric majorant in the weighted sum. Thus A_* lies beyond the reciprocal-summability criterion but satisfies (1). This is the source paper's ordinary calculation, with no separate entry in the paper-to-Lean ledger [2, calculation following the weighted proof].

A positive displacement. Suppose that $X_A(b) = p/q$ with integers p, q and $q > 0$. For a positive integer N , multiplication by $b^N - 1$ removes the complete geometric blocks from each summand, leaving

$$\Delta_{b,A}(N) = \sum_{a \in A} \frac{b^{N \bmod a} - 1}{b^a - 1} = (b^N - 1)X_A(b) - J_{b,A}(N), \quad N \geq 1, \quad (2)$$

where $J_{b,A}(N) = \sum_{a \in A, a \leq N} \sum_{j=1}^{\lfloor N/a \rfloor} b^{N-ja}$ is an integer. Every summand is nonnegative, and one is positive because the infinite set A contains an exponent larger than N . Hence $q\Delta_{b,A}(N)$ is a positive integer, and $\Delta_{b,A}(N) \geq 1/q$.

We obtain a contradiction by finding a multiple $N = tQ$ with smaller displacement. Divisibility helps because the summand indexed by a vanishes whenever $a \mid Q$. Choose a finite nonempty $F \subset A$ outside which the weighted sum is less than ε , and make Q a multiple of every element of F . Those terms then disappear for every t ; we must control the remaining terms on average.

A finite orbit and its unfinished end. With $b = 2$, $Q = 4$ and $a = 6$, the residues $tQ \bmod a$ are 4, 2, 0, and their mean displacement is

$$\frac{1}{3} \left(\frac{15}{63} + \frac{3}{63} + 0 \right) = \frac{2}{21} \leq \frac{1}{9}.$$

For a general exponent a , put $g = \gcd(a, Q)$. The residues run through the multiples of g in a cycle of length a/g . Summing the geometric progression over complete cycles, and bounding one unfinished cycle, gives

$$\frac{1}{T} \sum_{t=1}^T \frac{b^{tQ \bmod a} - 1}{b^a - 1} \leq \underbrace{\frac{g}{a(b^g - 1)}}_{\text{complete cycles}} + \underbrace{\frac{1}{T(b^g - 1)}}_{\text{unfinished cycle}}. \quad (3)$$

The first term has the shape of the weighted hypothesis once suitable prime powers have been put into Q . The second has lost the factor $1/a$. Summability of the first therefore gives no licence to sum the unfinished-cycle bounds over all exponents. This is the difficulty that a single progression average leaves unresolved.

A second average restores the missing factor. The source proof averages over $T = 2^j$, $M \leq j < 2M$, for a large integer M . For nonnegative weights α_a indexed by positive integers, with $\sum_a \alpha_a/a < \infty$, it uses

$$\sum_{j=M}^{2M-1} 2^{-j} \sum_{a \leq Q2^j} \alpha_a \leq 2Q \sum_a \frac{\alpha_a}{a}. \quad (4)$$

Indeed, a fixed a enters only when $2^j \geq a/Q$, and the sum of the admissible 2^{-j} is at most $2Q/a$. The missing factor is recovered by changing the length of the average, before summing over a .

To see why both averages can be made small, let $p_* = \max P$, take L divisible by F , and, for $H \geq 2p_*$, set

$$Q = L \prod_{p \in P} p^{\lfloor \log_p H \rfloor}, \quad G = \lfloor H/p_* \rfloor, \quad M = \lfloor b^{G/2} \rfloor.$$

If $h_p(a) \leq H$, then $h_p(a) \mid Q$; otherwise $\gcd(a, Q) \geq G$. Thus the small- $h_p(a)$ terms are controlled by the weighted sum and (4), while the large- $h_p(a)$ terms gain a denominator $b^G - 1$. With F and its common multiple L fixed, $Q \leq LH^{|P|}$, whereas M grows exponentially in H . Consequently $Q/M \rightarrow 0$ and $GM/b^G \rightarrow 0$. The finite mean is bounded by ε plus terms tending to zero; the exponents $a > QT$ contribute at most $4/T$. These are the two scale choices in the source proof [2, Section 2].

Since ε is arbitrary, one of the displacements is smaller than $1/q$, a contradiction. Passing to an infinite $B \subseteq A$ only decreases the weighted sum, and $b^{h_p(a)} - 1 \geq 2^{h_p(a)} - 1$ gives the all-base conclusion from the binary hypothesis [2, Section 2].

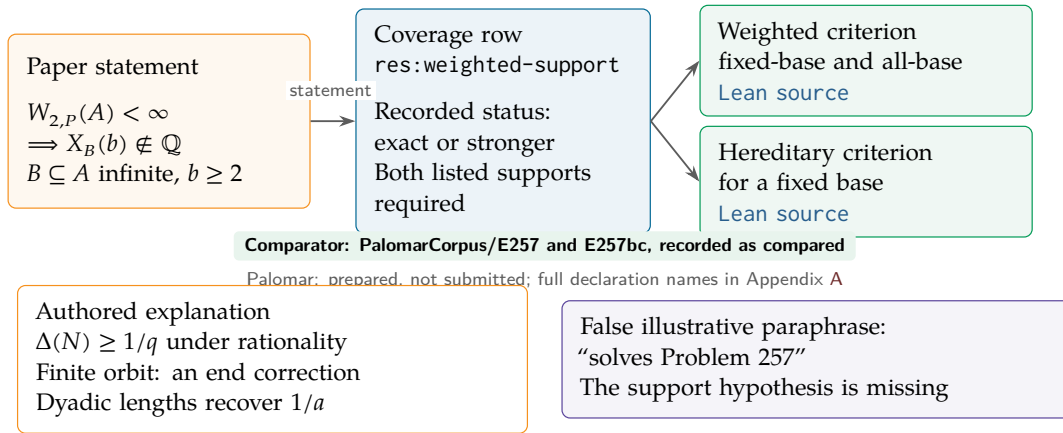


Figure 3: Both registered supports are required for the coverage row, while the authored explanation must justify the hypothesis and its use; omitting that hypothesis can change the claim without breaking either link, as the illustrative paraphrase shows (a different example from historical edit 8 in Table 1).

4 Proof and publication checks

Figure 3 now gives a concrete test for the publication checks. Its coverage record names two supporting Lean results. Evidence for the weighted criterion alone does not supply the complete registered set while the hereditary support is missing. The comparison includes every registered support and the additional hypotheses on which a conditional result depends, recorded as named inputs. It uses an identified source revision. A changed statement or source must be checked against its records again; a source file from one revision cannot silently borrow the evidence attached to another.

These checks concern different relations from the correctness of the formal proof and the meaning of the prose (Figure 4).

Relation	Lean	Comparator	Tools	Author	Review
Formal statement and proof	●				
Challenge and formal statement		●			
Sentence and declared source bytes			●		
Sentence and mathematical meaning				○	○
Result and motivating question				○	○
Novelty, significance and acceptance					○

Figure 4: Filled circles denote machine checks and open circles denote judgments; a source binding concerns declared bytes, while the relation between a sentence and its mathematical meaning remains a separate review question.

Lean verifies that a proof establishes the formal statement written in the source; it does not verify whether that statement captures the intended mathematics or whether the paper describes it well. Comparator adds a separately stated challenge and a check of its selected Lean implementation under the permitted axioms [19, 20].

For each prose passage we record its text, specified source lines, the kind of evidence and an author’s explanation of why those lines support the passage.

For example, the last paragraph of Section 3 attributes heredity to the decrease of a non-negative sum under restriction. Its record points to the weighted proof and explains that step. The checker compares cryptographic hashes of the text and source passage, and rejects missing records or passages without a binding.

That check detects an unrecorded edit. If an author changes both the sentence and its recorded explanation and updates the digests, the record becomes current again, but the claimed implication still needs review. In particular, every Lean link in Figure 3 can remain correct after the support condition has disappeared from an abstract. The reviewer must compare the hypothesis and quantifiers, not merely follow the links.

An entry in the declaration index with a matching type and a reachable source link identifies the recorded source; it does not establish that any particular continuous-integration run compiled the module. In the recorded design, the checking interface obtains each verdict from the specified Lean process, so callers cannot supply verdicts through that interface. This interface rule does not prevent an actor with filesystem access from changing the implementation or stored record. The protocol distinguishes contributor and reviewer roles but does not technically force a second independent mathematician.

5 From a Lean corpus to a paper

The worked example also suggests how to write the short account: give the support condition, show a set that it reaches, and explain why incomplete periods force a second average. A dependency graph locates the ingredients, but choosing this order and identifying the loss of $1/a$ are expository decisions made from the argument. The long record retains the complete proof and subsidiary cases. A correspondence link names the passage supporting a particular short-paper claim, so the reader can move from the explanation to the calculation without searching the whole record. The assembler preserves these authored parts and their cross-references; it supplies no missing mathematical step.

5.1 Source-frozen editorial refinement

A revision begins by collecting candidate results, antecedents, difficult proof transitions and unresolved questions with their source locations. The author decides which argument deserves the short paper; the checker audits source consistency and the specified manuscript rules. We freeze the selected manuscripts, input closure, coverage ledger and audit rules under a manifest of their exact bytes, so the writer and integrating reviewer can identify what the proposed changes were based on.

An advisory model can return a proposal against those sources; a tool-enabled agent can prepare it in a checkout. In either case the integrating reviewer decides whether the mathematics and its description remain faithful.

Before that decision, the return checker tests packet identity, permitted paths, statement-change declarations, labels and credit, and runs the manuscript audit on a temporary overlay of the proposed sources. It neither applies the proposal nor executes returned programs. Declaring a statement change makes the review obligation explicit; it does not discharge it.

The packet and audit stages in Figure 5 are implemented by `short_paper_writer.py`. Appendix A specifies the protocol roles and the acceptance steps. This procedure supplies no measurement of reader benefit or autonomous discovery.

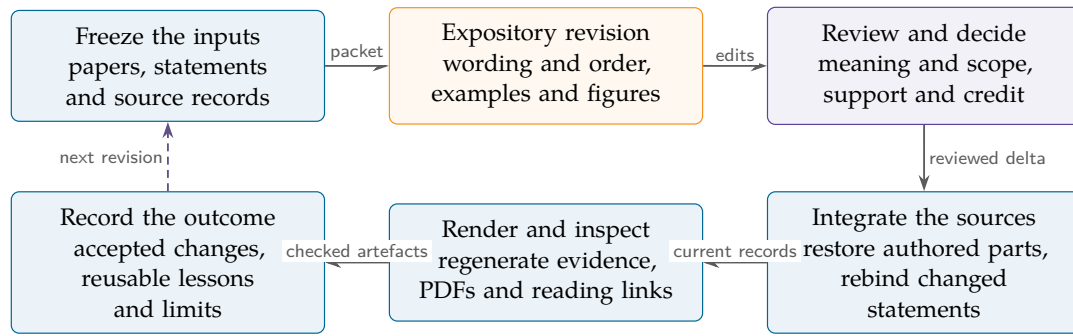


Figure 5: An exposition assignment returns focused source changes and optional figures for review; the stages describe responsibilities, not necessarily different people, and integration, rendering and the recorded outcome remain separate.

The relation to the motivating problem must also survive compression: the Problem 269 record distinguishes distinct running-LCM heights from the heights counted with multiplicity in the original question [28]. A result about the former is not automatically a result about the latter. When a statement is sharpened, merged or moved, its complete support set and any named input must move with it, and the statement relation needs renewed review. A new arrangement of a proof does not make its ingredients new.

6 An open route for contributions

A contributor may send a mathematical idea or reference without a clone, or return a patch from a recorded public commit. For executable work the return identifies the changed files, commands, results, resource use and surviving limitations, with the people and tools responsible. Figure 6 separates the proposed change from replay, review and adoption. We first reproduce an older branch at its original base and then reconcile it with current main, assigning separate credit to any substantive integration repair.

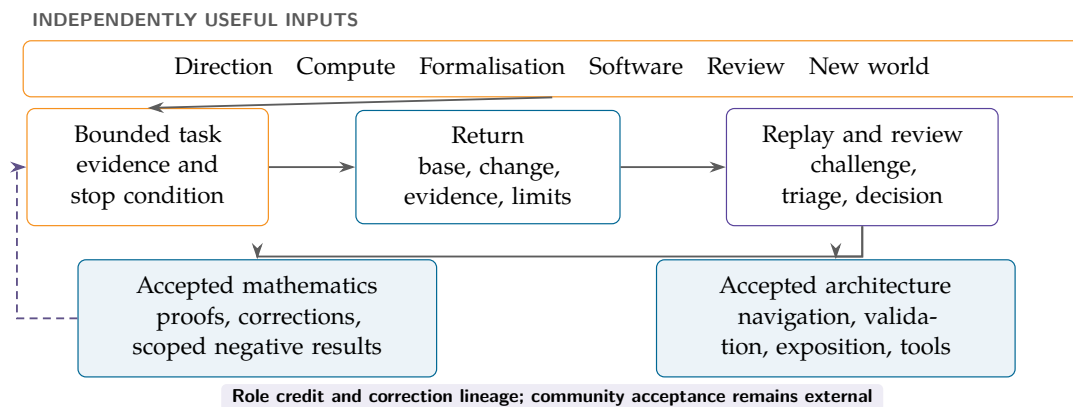


Figure 6: The contribution route accepts mathematical and architectural work under their own evidence classes; the feedback arrow records an adopted change and does not certify independent review or increased discovery rate.

The research journal records returns and dispositions in an append-only, hash-linked history, while a correction may withdraw something previously endorsed. Discovery investigates a question, while stewardship compares the return with prior work, groups related results, rec-

onciles affected papers and selects the next question. The proposed coupling is event-driven: a stable result or changed appraisal prompts work, whereas an unchanged repository supplies no reason for another run. The journal establishes what happened; judging what was learned requires the comparison with earlier work.

Generalising a local result requires a separate literature and library comparison, a proof of the proposed extension, and a second useful instance before an expert takes it to an upstream project [101]. That route is not automated and has produced no reported Mathlib contribution in the supplied record. We distinguish credit for an idea from credit for its implementation, formalisation or exposition, using CRediT as a role vocabulary rather than an authorship decision [105]. Later use adds an attributed dependency or correction, with neither a scalar impact score nor a financial contract inferred from that entry. A new world likewise needs a sourced question, explicit status, inspection routes and a review owner, which must still be assembled in several steps.

7 Recorded observations

The frozen inventory contains 689 paper occurrences, of which 640 have exact or specialising Lean support; 23 depend on a named input and 26 have no recorded Lean support. Of the 640 exact-or-specialising occurrences, 616 are recorded as compared and 24 as queued. These counts classify statement occurrences. Exact and specialising support match a paper statement directly or establish it by specialization. Named-input entries retain an additional assumed result; an entry without Lean support may still have an ordinary mathematical proof. Neither group is counted among the fully covered entries.

The historical publication-evidence record reports that nine of the ten deliberately false edits were rejected and one escaped. The edits were authored by the checker's author; the original run logs were not retained, and the other nine edits were not rerun against the extended checklist. The edits were applied separately, with the baseline restored between trials. The escaped edit changed a description of an unresolved step from "does not supply" to "completes", and the publication checks accepted it.

After the escape, the follow-up checked the intact baseline and that particular edit against an added check, leaving the other edits untested under the repair and supplying no post-repair ten-of-ten result. The post-repair witness accepts the current README and rejects a test copy containing the false clause. The sample comes from one corpus and one author's selection of errors, with neither a manual-review control nor an ordinary continuous-integration comparator. The study locates a coverage boundary. Its nine rejections do not estimate how reliable the checker is.

One architecture return was accepted on 23 September 2026 after focused checks of the ordinary-language return route and an architecture-packaging case. The contributor and maintainer-operated reviewer were the same agent; neither the receipt nor this paper claims an independent outside clone replay. As of 14 September 2026, no completed external cold-clone use had been recorded.

For readers, the proposed comparison holds the mathematical task fixed and contrasts an ordinary repository, an information-equivalent static briefing and the navigation tools. It would record understanding, severe errors and the costs of preparation, use, review and repair, including missing returns, timeouts and disagreements. No independent writing comparison, blind grading run or cold-reader experiment is reported here, and no comparative reader result is reported. General reliability, reader understanding and mathematical transfer remain unmeasured.

Edit	Change	First detector	Outcome
edit 1	Conditional result labelled proved	Projection freshness	rejected
edit 2	Open-boundary clause deleted	Boundary wording	rejected
edit 3	Declaration coordinate moved	Source coordinates	rejected
edit 4	Paper source link retargeted	Source coordinates	rejected
edit 5	Claim assigned to two families	Registry structure	rejected
edit 6	Module count hand-edited	Projection freshness	rejected
edit 7	native_decide introduced	Proof-trust policy	rejected
edit 8	“does not supply” changed to “completes”	None	escaped
edit 9	Open-problem title mangled	Paper anchoring	rejected
edit 10	Orientation byte budget exceeded	Byte budget	rejected

Historical trial: edit 8

The false completion clause passed all recorded checks.

Later witness: baseline and edit 8

Baseline accepted, edit 8 rejected; the other nine edits were not rerun.

Table 1: The ten historical edits and their recorded first detectors; “proof-trust policy” refers to this repository’s permitted proof mechanisms, not to a general unsoundness claim about native_decide.

8 Related work

The comparison with earlier systems is clearest at the relations they record. Lean and mathlib provide formal statements and proofs; blueprints link declarations to exposition; Prove2Me organises formalisation missions, reusable statements and contributions [11, 76, 35, 31]. The additional records here follow all registered supports and the source passages used in prose. Appendix B preserves the wider source map of the three original accounts. Prove2Me’s first arXiv version appeared on 28 August 2026, before the first public version of this paper, and already described public agent contributions, task decomposition, reusable statements and credit for reuse [31]. Its hosted verification service and this repository-based record provide different places for those activities; we claim neither priority for persistent research records nor a measured advantage over these systems. Polymath, BOINC and GIMPS supply precedents for small mathematical contributions and donated compute [100, 98, 99], while the Carleson blueprint shows formalisation organised into publicly claimable tasks [88].

The totient record refines Coons’s non- k -regularity antecedent and distinguishes Martin’s separation theorem from its own Chinese-remainder and Dirichlet argument traced to Yazdani [13, 14, 15, 16]. The running-LCM record retains Fan’s public priority for the two-prime case despite an independently reported derivation, and credits its Hecke–Mahler antecedents separately [28, 29, 26, 27]. For Problem 1041, the public degree-seven construction is ani’s; the recorded Lean result refutes the exact Formal Conjectures statement, while correspondence with the 1958 curve-length wording remains unreviewed [23, 77, 24, 22, 21]. Appendix A names its four recorded Lean declarations; Comparator is pending. The other seven targets remain unresolved here.

9 Limits

This paper reports no new Lean or Comparator execution, comparative formalisation result or autonomous-optimisation experiment. Transfer to unseen mathematics and understanding by

independent human readers are unresolved. The record supplies no controlled evidence that the contribution machinery improves the conversion from compute to mathematics or increases discovery rate. It also supplies no evidence that a new project can adopt the architecture cheaply or that an agent using it outperforms one using a good README. Full environment scans and initial builds still have costs, and an open corpus cannot establish that a model has never encountered its material.

The project remains maintainer-centred, with scarce review capacity and contestable credit; openness alone does not equalise resources or resolve conflicts and appeals. Broad mathematical acceptance is exogenous to this repository and cannot be granted by its maintainer. Returned code is untrusted, and expensive or privileged continuous-integration jobs must not execute fork code with repository secrets [106]. A volunteer-compute service and extensions to physical science remain proposals requiring their own validation, resource and safety governance.

The Advisory Group on Mathematics and Artificial Intelligence at IAS recommends distinguishing papers fully understood by a responsible mathematician from AI output that its human prompters do not understand, and asks AI laboratories to support subsequent community-led understanding [10]. Its 29 September 2026 statement gives concrete release expectations for attribution, exposition, formalisation and disclosure of the research process.

Several of these proposed release practices have counterparts in the records described here [10, Section 2.B, Step I]: the short paper and long record explain and attribute the argument, the paper-to-Lean ledger records formal correspondence and its limits, and the challenge files and generated formalization.yaml expose the comparison inputs. Those records make the result inspectable; they do not certify a mathematician’s understanding. The architecture alone does not establish complete per-result model, prompt, time and cost disclosure, an account of campaign-wide selection and unsuccessful attempts, or an independent scholarly deposit. The further recommendations on funding community-led understanding and equitable model access address AI laboratories; this prototype does not demonstrate those institutional outcomes.

The present contribution is an inspectable route through the records: a reader can recover the hypothesis, argument and support of a stated result, and identify where a source check ends and mathematical review begins.

A Inspection and reproduction

The initial unified manuscript was prepared from the frozen packet at `ced96e58b337`, a local integration commit rather than public main. Source links in this revision point to `7308896010e6b527d283dfdd0687a717a6183fb6`; the Comparator links below identify their separate recorded runs. The two companion manuscripts are retained as unchanged provenance for the unified account.

The weighted row in `docs/paper_lean_coverage.json` names `divisibilityWeightedClaim` and `finitePrimeWeighted_fixedBase_hereditary`, in namespace `ErdosProblems.Erdos257.PaperCompleteR8`. Its Comparator records are [run 35544127144](#) and [run 35624228171](#); the latter also records `PalomarCorpus/E257av` for the reciprocal antecedent. Both runs are recorded as compared; Palomar is prepared and not submitted.

For Erdős’s reciprocal-summability criterion, the formal declaration is `Erdos249257.irrational_erdosSupportSeries_of_summable_reciprocal`, with Comparator entry `PalomarCorpus/E257av` recorded as compared.

The four declarations supporting the reported 1041 refutation have namespace `Erdos1041`. Counterexample: `erdos1041_counterexample`, and the `HausdorffLength.lean` declarations `erdos1041_counterexample_hausdorff`, `erdos1041_hausdorff_negation` and `erdos1041_hausdorff_answer_false`. The coverage row gives exact-or-stronger Lean support, with Comparator pending.

In a complete checkout, `python3 scripts/systems_paper_evidence.py` checks this paper's source bindings, and `python3 scripts/build_systems_paper_counts.py` checks the generated inventory. The record at `docs/publication_evidence.json` holds the historical matrix and the limited follow-up, while `docs/publication_contract.json` declares the publication checks. Formal entry points include `Erdos249257.lean` and `.github/workflows/lean.yml`; the curated assertions are in `docs/claims.json`, and `scripts/check_release.py` runs the release gate. The editorial packet supplies selected Lean sources, rather than a complete buildable checkout; source-binding validation does not replay the Lean build, Comparator or the full repository release checks.

The protected snapshot partitions 689 occurrences into 498 exact, 142 specialising, 23 named-input and 26 unsupported-by-Lean entries. The journal summary at head `d3ad85269386` records 7 rounds (2 sealed), 25 returns (18 admitted and 7 unreviewed); earlier private-history rounds are not a complete public journal. The semantic counts in Figure 2 are the supplied interpretation inventory, not a count of individually reviewed proofs. Public reading begins with [the reading guide](#); `CONTRIBUTING.md` describes contribution channels under the existing [credit policy](#).

A.1 Editorial roles and acceptance

We call an agent that can alter the live repository Type A and an advisory agent returning proposals against supplied sources Type B.

This distinction concerns access to the repository. The assigned work is a separate choice: mathematical research, proof or claim audit, literature and attribution review, or exposition.

Research returns candidate arguments, an audit returns source-located findings, a literature review returns attribution proposals, and exposition returns a clearer manuscript; each product has its own review and acceptance criteria.

An exposition assignment preserves the established hypotheses, conclusions and evidence status; a suspected mathematical defect is returned for a separate audit or research assignment.

The reviewer compares hypotheses and implication directions with the whole statement and all its registered declarations. An ordinary argument retains its actual evidence class where formalisation is incomplete.

After acceptance, the long record's authored sections are restored, evidence and navigation are regenerated, and the compiled pages are inspected, including figures and proof links.

A writing rule learned from the revision travels with its example and limits as a candidate. Its adoption requires a reviewed change to the owning writing contract.

B Sources by role

Source family	Role in the three source accounts
Formal foundations	Lean and mathlib; library maintenance and contributor practice [11, 76, 93, 101]
Source and statement checks	Comparator, kernel replay, vacuity, linters and property tests [19, 20, 52, 43, 51]
Formal-to-prose tools	Blueprints, source documentation, translation and refactoring [35, 74, 37, 36, 61, 70]
Retrieval and interaction	Premises, proof states and exact declaration neighbourhoods [39, 89, 91, 92, 90, 94]
Proof automation	Premise selection and tactic search [49, 50, 59]
Library structure	Declaration networks and repository knowledge graphs [45, 46, 96]
Assumptions and reuse	Proof mining, generalisation, lemma mining and theory exploration [41, 42, 53, 54]
Growing proof knowledge	Library learning, evolving procedures and candidate statements [55, 56, 57, 58, 62, 84]
Research agents	Executable objectives, proof cost and evolving tasks [78, 79, 80, 30, 25, 60]
Agent collaboration	Bounties, planner/worker state, retrieval and compiler feedback [81, 82, 83, 31, 44, 75]
Evaluation	Research budgets, proof engineering and documentation tasks [85, 71, 72, 73, 107, 103, 104]
Validity and intended meaning	Research case studies, interactive proving and public accounting [18, 32, 40, 33]
Public infrastructure	Problem sources, formal conjectures, verification and collaboration [1, 77, 102, 88, 100, 98, 99, 105, 106]
Writing and maintenance	Mathematical exposition, build dependencies and provenance [86, 69, 48, 47, 95]
Understanding and reuse	Explanatory generality, digestion and research possibilities [5, 6, 9, 87, 7, 63, 97]
Different community views	Declarations, dissent, benchmarks, research infrastructure and pace [8, 64, 65, 66, 34, 67, 68]
Mathematical antecedents	Weighted support, totient ranks and arithmetic separation [2, 3, 4, 13, 14, 15, 16]
Other mathematical records	Running LCM, lemniscates and cross-problem connections [28, 29, 26, 27, 12, 38, 22, 21, 23, 24, 17]

Figure 7: Sources grouped by the roles for which the original accounts cite them, rather than by a performance ranking; the accompanying mathematical attribution distinguishes original results, refinements and independent rediscovery.

References

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